Questions by topic: Importance & features of energy resources - Set 1 (60 marks)

Answer all questions in the spaces provided.

0 1 Table 1 shows the per capita energy consumption and GDP per capita of five countries.

Table 1

Country	GDP per capita / US$ thousand	Energy use per capita / GJ per year	Main energy resource
A	62	290	Oil, gas
B	44	125	Gas, nuclear, renewables
C	10	95	Coal
D	2	25	Biomass (wood), oil
E	55	220	Hydroelectricity, oil

0 1 . 1 Describe the relationship between GDP per capita and energy use per capita shown in Table 1.

[2 marks]

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0 1 . 2 Explain why affluence increases per capita energy consumption.

[3 marks]

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0 1 . 3 Suggest **two** reasons why country B uses much less energy per capita than country A despite a similar level of affluence.

[2 marks]

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0 1 . 4 Explain why the per capita energy consumption of the UK has fallen in recent years even though consumption of goods has risen.

[2 marks]

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0 1 . 5 Give **one** way energy is used in each of the following: water supplies; agriculture.

[1 mark]

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0 2 Table 1 shows the approximate energy density of some energy resources.

Table 1

Energy resource	Energy density / MJ kg^{-1}
Uranium (natural, in fission reactor)	500 000
Natural gas	55
Petrol	46
Coal	28
Dry wood	16
Lithium-ion battery (stored electricity)	0.7

0 2 . 1 Explain what is meant by **energy density** and why it is an important feature of an energy resource.

[3 marks]

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0 2 . 2 Use Table 1 to calculate how many tonnes of coal contain the same energy as 1 kg of uranium.

[2 marks]

Answer ☐ tonnes

0 2 . 3 Explain why low energy density does not prevent solar and wind power from being useful.

[2 marks]

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0 2 . 4 Explain the difference between an **intermittent** resource and an **unpredictable** resource, using named examples.

[2 marks]

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0 2 . 5 Explain why natural gas must be liquefied for transport by ship.

[1 mark]

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0 3 . 1 Distinguish between **renewable** and **depletable** energy resources, giving an example of a resource that is both renewable and depletable.

[3 marks]

0 3 . 2 Explain why the abundance of an energy resource is not the same as the amount available for use. Give **two** examples.

[3 marks]

0 3 . 3 Describe the locational constraints of **two** named renewable energy resources.

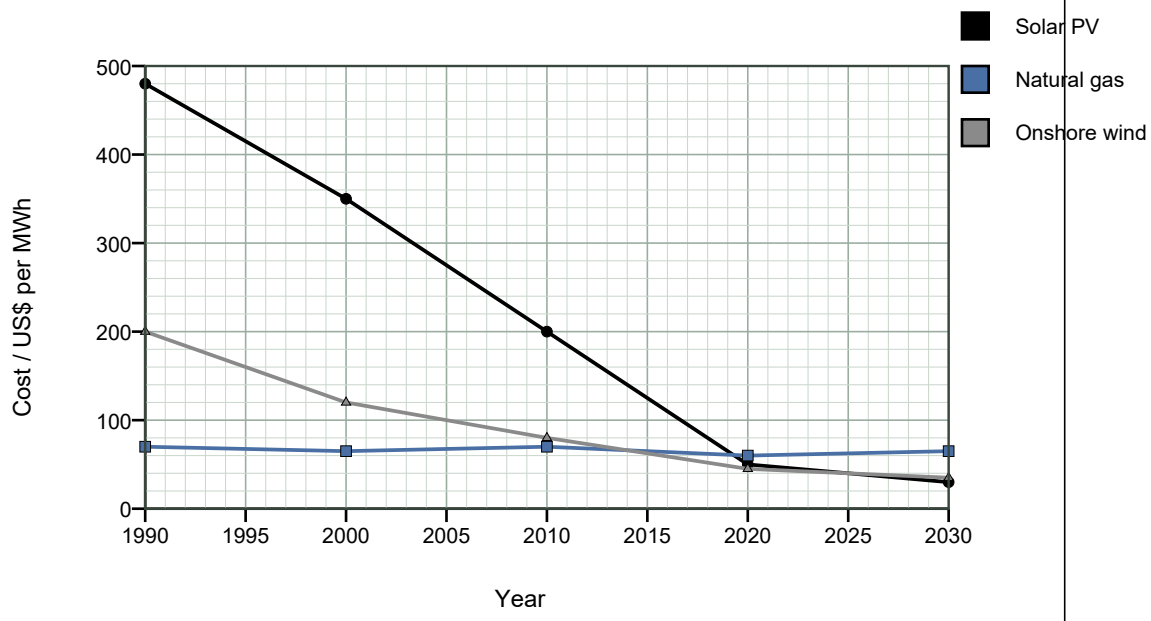
[4 marks]

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0 4 Figure 1 shows the estimated cost of generating electricity from three sources between 1990 and 2020, and projections to 2030.

Figure 1



0 4 . 1 Describe the trends shown in Figure 1.

[3 marks]

0 4 . 2 Explain why new energy technologies are expensive when first introduced and why their costs then fall.

[3 marks]

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0 4 . 3 Explain why a renewable energy project may still be at a financial disadvantage compared with a gas-fired power station of the same total cost.

[2 marks]

0 4 . 4 Explain why the price paid for fossil fuel energy does not reflect its full cost.

[2 marks]

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0 5 . 1 Explain why current global energy use is described as unsustainable.

[3 marks]

0 5 . 2 Describe the environmental impacts of exploiting fossil fuels that occur **before** the fuel is used.

[4 marks]

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0 5 . 3 Explain why an 'energy gap' could occur if society waits until fossil fuels become expensive before developing alternatives.

[3 marks]

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0 6 . 1 Discuss how the features of energy resources determine their suitability for different uses and their environmental sustainability.

[9 marks]

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0 6 . 2 State **one** reason why nuclear power stations can be located where transport infrastructure is poor.

[1 mark]

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END OF QUESTIONS